

Summary

- Final-year Electronics & Communication Engineering student specializing in Embedded Systems and Full-Stack IoT. Hands-on experience with embedded firmware development, PCB design (KiCad/Altium/EasyEDA), and integrating hardware with cloud dashboards. Proven track record in renewable energy research and practical network deployments.

Education

- **Dr. B. C. Roy Engineering College** Durgapur, West Bengal
B.Tech in Electronics & Communication Engineering (ECE); Final-year (Batch 2026) *Expected Jun. 2026*

Technical Skills

- **Embedded Firmware:** C/C++, Arduino Framework, ESP-IDF, Python (Scripting).
- **Hardware Design:** PCB Layout & Schematic (KiCad, Altium, EasyEDA), Circuit Analysis, Soldering & Rework.
- **Telecom & Networking:** 5G and FTTx deployments, failsafe design, telecom-grade routing & switching.
- **IoT & Cloud:** MQTT, HTTP/REST, Firebase/GCP, AWS, Azure, Docker.
- **Web Development:** JavaScript/TypeScript, Next.js, Express, React, PostgreSQL.
- **Lab Instruments:** Oscilloscope, Multimeter, Logic Analyzer, Bench Power Supply.

Experience

- **BSNL** Burdwan, West Bengal
Telecommunications Intern *Summer 2025*
 - **Routing & Switching:** Analyzed telecom-grade routing and switching protocols used in live networks.
 - **Field Support:** Assisted field engineers in troubleshooting signal integrity issues for fiber optic endpoints.
- **Tata Power SDI (PMSGY)** Maithon, Jharkhand
Supervising Engineer Trainee *Winter 2026*
 - **Training Focus:** Supervised solar PV cell installation, safety compliance, and on-site commissioning.
 - **Site Work:** Assisted in PV site surveys, panel mounting, and safety inspections.

Projects

- **Implementation of 800W MPPT using ESP32 for Rural Electrification** Academic
ESP32 • MPPT • Incremental Conductance *Winter 2026*
 - **Overview:** Designed and implemented an 800W synchronous MPPT solar charge controller to improve conversion efficiency for rural electrification and smart energy management.
- **Intelligent Smart Grid & Energy Telemetry Platform** Academic
Edge Computing • Power Metrology • Real-Time Cloud DB • Web Framework *Fall 2025*
 - **Overview:** Microcontroller-driven prepaid energy meter streaming real-time telemetry to a secure cloud database and a responsive web dashboard.
 - **Key Results:** Implemented high-precision metering and remote relay cutoff with local response < 200ms.

Research & Publications

- **Weather-Integrated PEMS for Residential Hybrid Solar Installations** Electrica journal
Under Review (Submission ID: electr-2025-0424) *Winter 2025*
 - **Overview:** Predictive energy management for residential hybrid solar using weather inputs and smart metering to improve scheduling and efficiency.
- **ESP32-Based Synchronous MPPT SCC with INC Algorithm** ICICET 2026
Accepted for Presentation (Speaker Category, Track 7) *Aug. 2026*
 - **Overview:** Abstract accepted for presentation at Bolpur, Shantiniketan. Demonstrated high conversion efficiency and strong applicability in renewable energy systems for rural electrification.

Interests

- Audio stories; DIY electronics projects; Solar & renewable tech; Reading research papers; Open-source; Fixing family electronics; Working on: <https://toolhub.live>